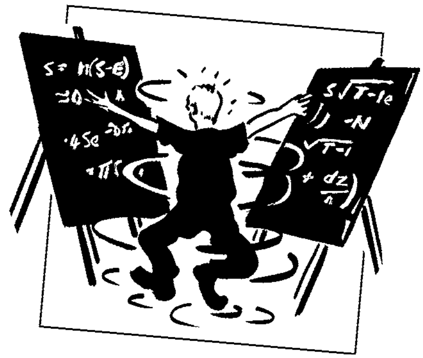
CHAPTER 7

partial differential equations



The aim of this Chapter...

...is to compare the Black–Scholes equation with mathematical models in other walks of life and so instill in the reader confidence in the relevance of partial differential equations, and to demonstrate some of the more useful solution methods...although we won't really be needing any of them.

In this Chapter...

- properties of the parabolic partial differential equation
- the meaning of terms in the Black–Scholes equation
- some solution techniques

7.1 INTRODUCTION

The analysis and solution of partial differential equations is a BIG subject. We can only skim the surface in this book. If you don't feel comfortable with the subject, then the list of books at the end should be of help. However, to understand finance, and even to solve partial differential equations numerically, does not require any great depth of understanding. The aim of this chapter is to give just enough background to the subject to permit any reasonably numerate person to follow the rest of the book; I want to keep the entry requirements to the subject as low as possible.

7.2 PUTTING THE BLACK–SCHOLES EQUATION INTO HISTORICAL PERSPECTIVE

The Black–Scholes partial differential equation is in two dimensions, S and t. It is a parabolic equation, meaning that it has a second derivative with respect to one variable, S, and a first derivative with respect to the other, t. Equations of this form are more colloquially known as **heat** or **diffusion equations**.

The equation, in its simplest form, goes back to almost the beginning of the 19th century. Diffusion equations have been successfully used to model

- diffusion of one material within another, smoke particles in air;
- flow of heat from one part of an object to another;
- chemical reactions, such as the Belousov–Zhabotinsky reaction which exhibits fascinating wave structure;
- electrical activity in the membranes of living organisms, the Hodgkin–Huxley model;
- dispersion of populations, individuals move both randomly and to avoid overcrowding;
- pursuit and evasion in predator–prey systems;
- pattern formation in animal coats, the formation of zebra stripes;
- dispersion of pollutants in a running stream.

In most of these cases the resulting equations are more complicated than the Black–Scholes equation.

The simplest heat equation for the temperature in a bar is usually written in the form

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

where u is the temperature, x is a spatial coordinate and t is time. This equation comes from a heat balance. Consider the flow into and out of a small section of the bar. The flow of heat along the bar is proportional to the spatial gradient of the temperature

$$\frac{\partial u}{\partial x}$$

and thus the derivative of this, the *second* derivative of the temperature, is the heat retained by the small section. This retained heat is seen as a rise in the temperature, represented mathematically by

$$\frac{\partial u}{\partial t}.$$

The balance of the second x-derivative and the first t-derivative results in the heat equation. (There would be a coefficient in the equation, depending on the properties of the bar, but I have set this to one.)

7.3 THE MEANING OF THE TERMS IN THE BLACK–SCHOLES EQUATION

The Black–Scholes equation can be accurately interpreted as a reaction-convection-diffusion equation. The basic diffusion equation is a balance of a first-order t derivative and a second-order S derivative:

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}.$$

If these were the only terms in the Black–Scholes equation it would still exhibit the smoothing-out effect, that any discontinuities in the payoff would be instantly diffused away. The only difference between these terms and the terms as they appear in the basic heat or diffusion equation is that the diffusion coefficient is a function of one of the variables S. Thus we really have diffusion in a non-homogeneous medium.

The first-order S-derivative term

$$rS\frac{\partial V}{\partial S}$$

can be thought of as a convection term. If this equation represented some physical system, such as the diffusion of smoke particles in the atmosphere, then the convective term would be due to a breeze, blowing the smoke in a preferred direction.

The final term

$$-rV$$

is a reaction term. Balancing this term and the time derivative would give a model for decay of a radioactive body, with the half-life being related to r. (A better description, for which I am indebted to a delegate on a training course but whose name I've forgotten, is that this term is a 'passive smoking' effect.)

Putting these terms together and we get a reaction-convection-diffusion equation. An almost identical equation would be arrived at for the dispersion of pollutant along a flowing river with absorption by the sand. In this, the dispersion is the diffusion, the flow is the convection, and the absorption is the reaction.

7.4 BOUNDARY AND INITIAL/FINAL CONDITIONS

To uniquely specify a problem we must prescribe **boundary conditions** and an **initial** or **final condition**. Boundary conditions tell us how the solution must behave for all time at certain values of the asset. In financial problems we usually specify the behavior of the solution at $S = 0$ and as $S \to \infty$. We must also tell the problem how the solution begins. The Black–Scholes equation is a backward equation, meaning that the signs of the t derivative and the second S derivative in the equation are the same when written on the same side of the equals sign. We therefore have to impose a final condition. This is usually the payoff function at expiry.

The Black–Scholes equation in its basic form is linear and satisfies the superposition principle; add together two solutions of the equation and you will get a third. This is not true of non-linear equations. Linear diffusion equations have some very nice properties. Even if we start out with a discontinuity in the final data, due to a discontinuity in the payoff, this *immediately* gets smoothed out; this is due to the diffusive nature of the equation. Another nice property is the uniqueness of the solution. Provided that the solution is not allowed to grow too fast as S tends to infinity the solution will be unique. This precise definition of 'too fast' need not worry us, we will not have to worry about uniqueness for any problems we encounter.

7.5 SOME SOLUTION METHODS

We are not going to spend much time on the exact solution of the Black–Scholes equation. Such solution is important, but current market practice is such that models have features which preclude the exact solution. The few explicit, closed-form solutions that are used by practitioners will be covered in the next chapter.

Time Out...

Do I need to know this?

No. You probably won't need to find explicit solutions in practice. Indeed, very rarely can explicit solutions be found to realistic financial problems. That's why I focus more on numerical methods in this book. Unless you are doing a heavily math orientated course, you can safely skip the rest of this chapter.

7.5.1 Transformation to constant coefficient diffusion equation

It can sometimes be useful to transform the basic Black–Scholes equation into something a little bit simpler by a change of variables. If we write

$$V(S,t) = e^{\alpha x + \beta \tau} U(x,\tau),$$

where

$$\alpha = -\tfrac{1}{2}\left(\frac{2r}{\sigma^2} - 1\right), \quad \beta = -\tfrac{1}{4}\left(\frac{2r}{\sigma^2} + 1\right)^2, \quad S = e^x \quad \text{and} \quad t = T - \frac{2\tau}{\sigma^2},$$

then $U(x, \tau)$ satisfies the basic diffusion equation

$$\frac{\partial U}{\partial \tau} = \frac{\partial^2 U}{\partial x^2}. \tag{7.1}$$

This simpler equation is easier to handle than the Black–Scholes equation. Sometimes that can be important, for example when seeking closed-form solutions, or in some simple numerical schemes. We shall not pursue this any further.

7.5.2 Green's functions

One solution of the Black–Scholes equation is

$$V'(S,t) = \frac{e^{-r(T-t)}}{\sigma S' \sqrt{2\pi(T-t)}} e^{-\left(\log(S/S') + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2 (T-t)} \tag{7.2}$$

for any S'. (You can verify this by substituting back into the equation, but we'll also be seeing it derived in the next chapter.) This solution is special because as $t \to T$ it becomes zero everywhere, except at $S = S'$. In this limit the function becomes what is known as a **Dirac delta function**. Think of this as a function that is zero everywhere except at one point where it is infinite, in such a way that its integral is one. How is this of help to us?

Expression (7.2) is a solution of the Black–Scholes equation for any S'. Because of the linearity of the equation we can multiply (7.2) by any constant, and we get another solution. But then we can also get another solution by adding together expressions of the form (7.2) but with different values for S'. Putting this together, and thinking of an integral as just a way of adding together many solutions, we find that

$$\frac{e^{-r(T-t)}}{\sigma \sqrt{2\pi(T-t)}} \int_0^\infty e^{-\left(\log(S/S') + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2 (T-t)} f(S') \frac{dS'}{S'}$$

is also a solution of the Black–Scholes equation for any function $f(S')$. (If you don't believe me, substitute it into the Black–Scholes equation.)

Because of the nature of the integrand as $t \to T$ (i.e. that it is zero everywhere except at S' and has integral one), if we choose the arbitrary function $f(S')$ to be the payoff function then this expression becomes the solution of the problem:

$$V(S,t) = \frac{e^{-r(T-t)}}{\sigma \sqrt{2\pi(T-t)}} \int_0^\infty e^{-\left(\log(S/S') + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right)^2 / 2\sigma^2 (T-t)} \text{Payoff}(S') \frac{dS'}{S'}.$$

The function $V'(S,t)$ given by (7.2) is called the **Green's function**.

7.5.3 Series solution

Sometimes we have boundary conditions at two finite (and non-zero) values of S, S_u and S_d, say (we see examples in Chapter 13). For this type of problem, we postulate that the required solution of the Black–Scholes equation can be written as an infinite sum of special functions. First of all, transform to the nicer basic diffusion equation in x and τ. Now write the solution as

$$e^{\alpha x+\beta\tau}\sum_{i=0}^{\infty} a_i(\tau)\sin(i\omega x)+b_i(\tau)\cos(i\omega x),$$

for some ω and some functions a and b to be found. The linearity of the equation suggests that a sum of solutions might be appropriate. If this is to satisfy the Black–Scholes equation then we must have

$$\frac{da_i}{d\tau}=-i^2\omega^2 a_i(\tau) \quad\text{and}\quad \frac{db_i}{d\tau}=-i^2\omega^2 b_i(\tau).$$

You can easily show this by substitution. The solutions are thus

$$a_i(\tau)=A_i e^{-i^2\omega^2\tau} \quad\text{and}\quad b_i(\tau)=B_i e^{-i^2\omega^2\tau}.$$

The solution of the Black–Scholes equation is therefore

$$e^{\alpha x+\beta\tau}\sum_{i=0}^{\infty} e^{-i^2\omega^2\tau}(A_i\sin(i\omega x)+B_i\cos(i\omega x)). \tag{7.3}$$

We have solved the equation, all that we need to do now is to satisfy boundary and initial conditions.

Consider the example where the payoff at time $\tau=0$ is $f(x)$ (although it would be expressed in the original variables, of course) but the contract becomes worthless if ever $x=x_d$ or $x=x_u$.[1]

Rewrite the term in brackets in (7.3) as

$$C_i\sin\left(i\omega'\frac{x-x_d}{x_u-x_d}\right)+D_i\cos\left(i\omega'\frac{x-x_d}{x_u-x_d}\right).$$

To ensure that the option is worthless on these two x values, choose $D_i=0$ and $\omega'=\pi$. The boundary conditions are thereby satisfied. All that remains is to choose the C_i to satisfy the final condition:

$$e^{\alpha x}\sum_{i=0}^{\infty} C_i\sin\left(i\omega'\frac{x-x_d}{x_u-x_d}\right)=f(x).$$

This also is simple. Multiplying both sides by

$$\sin\left(j\omega'\frac{x-x_d}{x_u-x_d}\right),$$

and integrating between x_d and x_u we find that

$$C_j=\frac{2}{x_u-x_d}\int_{x_d}^{x_u} f(x)e^{-\alpha x}\sin\left(j\omega'\frac{x-x_d}{x_u-x_d}\right)dx.$$

This technique, which can be generalized, is the **Fourier series method**. There are some problems with the method if you are trying to represent a discontinuous function with a sum of trigonometrical functions. The oscillatory nature of an approximate solution with a finite number of terms is known as **Gibbs phenomenon**.

[1] This is an example of a double knock-out option, see Chapter 13.

7.6 SIMILARITY REDUCTIONS

Apart from the Green's function, we're not going to use any of the above techniques in this book; rarely will we even find explicit solutions. But one technique that we will find useful is the **similarity reduction**. I will demonstrate the idea using the simple diffusion equation, we will later use it in many other, more complicated problems.

The basic diffusion equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \tag{7.4}$$

is an equation for the function u which depends on the two variables x and t. Sometimes, in very, very special cases, we can write the solution as a function of just one variable. Let me give an example. Verify that the function

$$u(x,t) = \int_0^{x/t^{1/2}} e^{-\frac{1}{4}\xi^2}\, d\xi$$

satisfies (7.4). But in this function x and t only appear in the combination

$$\frac{x}{t^{1/2}}.$$

Thus, in a sense, u is a function of only one variable.

A slight generalization, but also demonstrating the idea of similarity solutions, is to look for a solution of the form

$$u = t^{-1/2} f(\xi) \tag{7.5}$$

where

$$\xi = \frac{x}{t^{1/2}}.$$

Substitute (7.5) into (7.4) to find that a solution for f is

$$f = e^{-\frac{1}{4}\xi^2},$$

so that

$$t^{-1/2} e^{-\frac{1}{4}\frac{x^2}{t}}$$

is also a special solution of the diffusion equation.

Be warned, though. You can't always find similarity solutions; not only must the equation have a particularly nice structure but also the similarity form must be consistent with any initial condition or boundary conditions.

7.7 OTHER ANALYTICAL TECHNIQUES

The other two main solution techniques for linear partial differential equations are Fourier and Laplace transforms. These are such large and highly technical subjects that I really cannot begin to give an idea of how they work, space is far too short. But be reassured that it is probably not worth your while learning the techniques, since in finance they can be used to solve only a very small number of problems. If you want to learn something useful then move on to the next section.

7.8 NUMERICAL SOLUTION

Even though there are several techniques that we can use for finding solutions, in the vast majority of cases we must solve the Black–Scholes equation numerically. But we are lucky. Parabolic differential equations are just about the easiest equations to solve numerically. Obviously, there are any number of really sophisticated techniques, but if you stick with the simplest then you can't go far wrong. I want to stress that I am going to derive many partial differential equations from now on, and I am going to assume you trust me that we will at the end of the book see how to solve them.

7.9 SUMMARY

This short chapter is only intended as a primer on partial differential equations. If you want to study this subject in depth, see the books and articles mentioned below.

FURTHER READING

- Grindrod (1991) is all about reaction-diffusion equations, where they come from and their analysis. The book includes many of the physical models described above.
- Murray (1989) also contains a great deal on reaction-diffusion equations, but concentrating on models of biological systems.
- Wilmott & Wilmott (1990) describe the diffusion of pollutant along a river with convection and absorption by the river bed.
- The classical reference works for diffusion equations are Crank (1989) and Carslaw & Jaeger (1989). But also see the book on partial differential equations by Sneddon (1957) and the book on general applied mathematical methods by Strang (1986).

Time Out...

The main solution methods

We have seen, and will be seeing more of, the three main mathematical approaches to derivative pricing: differential equations; binomial trees; expectations. All of these methods are based on pretty much the same assumptions. All of them will therefore give the same values for a contract, if all parameter values are the same. This is, of course, subject to the accuracy of numerical methods.

Speaking of numerical methods, each of the three approaches has its own associated numerical method. Differential equations, and the Black–Scholes equation, in particular, can be solved by finite-difference methods. The whole of Chapter 28 is devoted to this subject. The binomial tree model is, interestingly, also its own numerical method, and we've seen that in some detail already in Chapter 3. Finally, pricing by calculating risk-neutral expectations is one of the subjects in Chapter 29 on Monte Carlo simulations.

EXERCISES

1. Consider an option with value $V(S, t)$, which has payoff at time T. Reduce the Black–Scholes equation, with final and boundary conditions, to the diffusion equation, using the following transformations:

   (a)

   $$S = Ee^x, \quad t = T - \frac{2\tau}{\sigma^2}, \quad V(S, t) = Ev(x, \tau),$$

   (b)

   $$v = e^{\alpha x + \beta\tau} u(x, \tau),$$

   for some α and β. What is the transformed payoff? What are the new initial and boundary conditions? Illustrate with a vanilla European call option.

2. The solution to the initial value problem for the diffusion equation is unique (given certain constraints on the behavior, it must be sufficiently smooth and decay sufficiently fast at infinity). This can be shown as follows:

   Suppose that there are two solutions $u_1(x, \tau)$ and $u_2(x, \tau)$ to the problem

   $$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{on } -\infty < x < \infty,$$

   with

   $$u(x, 0) = u_0(x).$$

   Set $v(x, \tau) = u_1 - u_2$. This is a solution of the equation with $v(x, 0) = 0$. Consider

   $$E(\tau) = \int_{-\infty}^{\infty} v^2(x, \tau)\, dx.$$

   Show that

   $$E(\tau) \geq 0, \quad E(0) = 0,$$

   and integrate by parts to find that

   $$\frac{dE}{d\tau} \leq 0.$$

   Hence show that $E(\tau) \equiv 0$ and, consequently, $u_1(x, \tau) \equiv u_2(x, \tau)$.

3. Suppose that $u(x, \tau)$ satisfies the following initial value problem:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{on } -\pi < x < \pi, \quad \tau > 0,$$

with

$$u(-\pi, \tau) = u(\pi, \tau) = 0, \quad u(x, 0) = u_0(x).$$

Solve for u using a Fourier sine series in x, with coefficients depending on τ.

4. Check that u_δ satisfies the diffusion equation, where

$$u_\delta = \frac{1}{2\sqrt{\pi\tau}} e^{-\frac{x^2}{4\tau}}.$$

5. Solve the following initial value problem for $u(x, \tau)$ on a semi-infinite interval, using a Green's function:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{on } x > 0, \quad \tau > 0,$$

with

$$u(x, 0) = u_0(x) \text{ for } x > 0, \quad u(0, \tau) = 0 \text{ for } \tau > 0.$$

Hint: Define $v(x, \tau)$ as

$$v(x, \tau) = u(x, \tau) \text{ if } x > 0,$$
$$v(x, \tau) = -u(-x, \tau) \text{ if } x < 0.$$

Then we can show that $v(0, \tau) = 0$ and

$$u(x, \tau) = \frac{1}{2\sqrt{\pi\tau}} \int_0^\infty u_0(s)(e^{-(x-s)^2/4\tau} - e^{-(x+s)^2/4\tau})\, ds.$$

6. Reduce the following parabolic equation to the diffusion equation.

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2} + a\frac{\partial u}{\partial x} + b,$$

where a and b are constants.

7. Using a change of time variable, reduce

$$c(\tau)\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2},$$

to the diffusion equation when $c(\tau) > 0$.

Consider the Black–Scholes equation, when σ and r can be functions of time, but $k = 2r/\sigma^2$ is still a constant. Reduce the Black–Scholes equation to the diffusion equation in this case.

8. Show that if

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{on } -\infty < x < \infty, \quad \tau > 0,$$

with

$$u(x, 0) = u_0(x) > 0,$$

then $u(x, \tau) > 0$ for all τ.

Use this result to show that an option with positive payoff will always have a positive value.

9. If $f(x, \tau) \geq 0$ in the initial value problem

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2} + f(x, \tau), \quad \text{on } -\infty < x < \infty, \quad \tau > 0,$$

with

$$u(x, 0) = 0, \text{ and } u \to 0 \text{ as } |x| \to \infty,$$

then $u(x, \tau) \geq 0$. Hence show that if C_1 and C_2 are European calls with volatilities σ_1 and σ_2 respectively, but are otherwise identical, then $C_1 > C_2$ if $\sigma_1 > \sigma_2$.

Use put-call parity to show that the same is true for European puts.